Event Management — Exercise

Frontend Developer (Angular / DevExtreme)

English translation of the German exercise PDF

Background

Our Corporate Marketing team organises a large number of events every year. These include trade fairs, congresses and customer events, but also internal events such as the summer party or the Christmas party. Some of these events recur every year, others take place only once. An event can have up to several thousand participants.

So far this has been handled with Excel lists. The colleagues need a web application that supports them in organising these events. It should be fast, clear and usable without unnecessary clicks.

What The Corporate Marketing Team Needs

- Keep an overview of all events: date, location, status (planned, cancelled, done) and number of participants - smoothly at all times, even with more than 5,000 entries.
- Find things quickly: narrow down by status and location, by time period, and by free text across all fields. Settings, once made, should still be there after a reload.
- Analyse: see participant numbers summarised by status or by location.
- Maintain: create new events (the date must be in the future; required fields). Change the status and location of an existing event quickly, without leaving the overview.
- The possible statuses will change and should in future come from the backend, not be hard-coded.
- View, add and remove the participants of each event (name, email, role) - also without leaving the overview. Participant data should only be loaded when it is really needed.
- Share data: export of the current view. The export goes to internal or external people. With German date format, and with the event name and the export date in the header.
- Over the next months the application will grow to include further areas (for example venues, invoices) with similar list requirements. The structure should support this without reinventing every list.

Data

No real backend. Simulate a service with at least 5,000 events (3-10 participants per event), 100-300 ms latency and occasional errors - for example every twentieth request fails. Paging, sorting and filtering should be implemented as if a real backend were behind them.

Technical Framework

- Angular 19 or newer, TypeScript with strict typing.
- UI components: DevExtreme.
- Everything else - project structure, component split, state management, data access, test strategy - is up to you. These decisions interest us at least as much as the result.
- German UI texts, accessibility within reason.

Submission

- Runnable project as a GitHub link or zip file.
- README with:
  - setup instructions;
  - a short architecture overview;
  - a decision log - at least three significant decisions, each with the alternatives you considered and the reasoning;
  - assumptions you made where requirements were unclear;
  - how you achieved smooth performance with 5,000+ entries and how you verified it;
  - known limitations;
  - what you would do differently or additionally with more time.
- Tests where you consider them important - with a short justification of your selection.
- Guideline: one working day. What you leave out when time is short, and why, is part of the submission.
- Send it to us one working day before the appointment.

At The Appointment

In the first 15 minutes you present the architecture and decisions. After that we discuss your solution together.